CS496 Software Project:
Web-Based HVAC Inventory App

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1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name: William Tikiob
- Client title: Project Manager Engineer
- Client email address: william.tikiob@coolsys.com
- Client employer: CoolSys
- How you know the client: Introduced to by Dr. Isaacman

2 Project Description

2.1 Overview

The main priority of this software, and the problem we are trying to solve, is the tracking of materials for jobs. Tracking materials is divided among two roles. The project managers order the needed materials and keep track of the purchase orders (PO's) in files on a shared drive. At the job sites, the foremen receive the materials and sign off on the delivery. Currently, there is no way for the foremen to verify that all of the materials were actually shipped, which has lead to delays because of missing things.

Our app would centralize the ordered materials in a place for everyone in the company to see. They would be organized by job, so comparing expected and delivered materials would be as simple as finding the appropriate job on the app. Project managers would be able to take photos of PO's, and - possibly using Optical Character Recognition (OCR) - the materials and their quantities would be inserted into a database. In addition, the people on site for these jobs should be able to take a photo of the packing slip, and the app would compare that list to the expected numbers in the database.

Those on job sites will require an easy-to-use mobile app. It makes sense to make a system that will work both on a computer and on mobile devices.

If the former is complete, there are secondary additions that the client requested. These are equipment serial capture and tool tracking. Tool tracking would allow all users to know what tools are currently allocated to certain people/jobs. Serial capture would allow the automation of inserting these tools into a DB.

2.2 Key Features

- Inserting material list into central DB (Project Manager)
- Organizing materials by job

- Comparing delivered materials to expected material list (Foremen)
- Automating insertion and comparison, possibly via OCR (client's recommendation)
- Custom role-based UI's (the simpler, the better)
- Mobile App design
- (stretch goal) Tracking tools by job and/or person
- (stretch goal) Equipment serial capture

2.3 Why this Project is Interesting

(Oscar) I like this project because of the high-level simplicity. The material lists are the primary data. All we want to do is provide a way for project managers to upload the lists and organize them, and for the foremen to review them. Obviously, it will be more difficult to implement than how I described it, but dealing with the logic of only one major data type is something that I am prepared to deal with. I also understand the perspective of the foremen and the confusion and anger that comes with incomplete orders on job sites. Being able to help their work run more smoothly is a bonus.

2.4 Areas of CS required

[What subfields of computer science seem most likely to be relevant to your project? A capstone must involve multiple.]

1. Software Engineering
2. Web/Mobile App Development
3. DB Management
4. (possibly) Optical Character Recognition

2.5 Potential Concerns and Questions

It is unclear whether or not we will be able to get to the secondary additions to the app. Going into the project, it seems that organizing and tracking tools should be similar to the material lists. Tool tracking could be moved from being a stretch goal to part of the main requirements.

One concern we have is integrating OCR into the project. The company offered to pay for AWS Textract, which (according to them) has the necessary capabilities. This is interesting, but we are unsure of how this contributes to the difficulty of the project. While this is a concern, the focus at first should be on the presentation of the data.

There are also a few questions we have, both for the client and in general:

1. How would we get OCR to handle different PO formats, if necessary?
2. How should we differentiate tools being used *at a job site* versus *by a specific person*?
3. Should admin create accounts or should users create their own accounts?
4. Who manages the tools (admin, pm, somebody else)?
5. How do you handle delivery status?

3 Requirements

3.1 Non-Functional Requirements

Table 1 presents the NFRs for this software project.

ID	NFR Title	Category	Description
1	Color Scheme	Visual / Design	No specifc colors required, but primary blue and red to match company colors
2	Simple UI	Visual / Design	The simpler the UI, the better
3	Role-specific UI	Portability	Different UI for different roles
4	Mobile App	Portability	Mobile version for foremen
2	OCR Handling	Reliability	OCR should account for different PO/Packing slip formats

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

Table 2 presents the functional requirements for this software project.

ID	Story Title	Points	Description
A1	User Accounts	8	As a/an Admin, I want to create, update, and delete user accounts, so that each user has access to needed information
P1	Material Lists	3	As a/an Project Manager, I want to upload material lists based on PO, so that users can see what is needed for projects
P2	PO Photo	3	As a/an Project Manager, I want to upload a photo of a purchase order, so that I can avoid tidious input and errors
D1	Projects	5	As a/an Project Manager, I want to create, update, and delete projects, so that I can keep projects up to date
D2	Organize Material	3	As a/an Project Manager, I want to separate material lists by project, so that lists are organized
D3	Assign Users	2	As a/an Project Manager, I want to assign users to specific projects, so that they are not distracted by other projects
D4	Material List Status	5	As a/an Project Manager, I want to see the status of material deliveries across all projects, so that I can easily keep track of accuarcy and progress
D5	Shipment Notification	2	As a/an Foreman/Logistics, I want a notification when materials are ordered, so that I know when to expect them at a project site
T1	Tool Management	8	As a/an Foreman, I want to create, read, update, and delete company tools, so that I can manage them
D6	Assign/Unassign Tools	3	As a/an Foreman, I want to assign tools to specific projects/people, so that users know what tools are in use and where
S1	Record Shipment	2	As a/an Logistics, I want to input delivered materials , so that I can create a shipment and read a PO to compare for accuacy
S2	Shipment Accuarcy	5	As a/an Logistics, I want to compare packing lists to material lists, so that I know if any materials are missing
S3	Shipment Photo Upload	3	As a/an Logistics, I want to upload a photo of a packing slip, so that delivered materials can be automatically recorded
D7	Tool Allocation	2	As a/an Foreman/Logistics, I want to see my tool when I am assigned to a project, so that other users know where a specific tool is
Q1	Project Details on Forms	2	As a/an Quality Control, I want documentation to populate with project details, so that time is saved and error is minimized on projects
Q2	Equipment Serial Capture	5	As a/an Quality Control, I want to take a photo of a serial nameplate and input its details onto QC form, so that time is saved and error is minimized on projects

M1	Persistent Login and Logout	3	As a/an User, I want to login, remain logged in, and logout of my account, so that my information is secure and maintained
D8	User Dashboard	3	As a/an User, I want to see my info (role, tools, projects) on my profile so that I have all of my personal info available
P3	Project Popup	3	As a/an Project Manager, I want a popup of a project that I search, so that I can quickly see the information I want without other projects cluttering the screen
T2	Tool Popup	3	As a/an Foreman/Logistics, I want a popup of the tool that I search, so that I can quickly see the information I want without other tools cluttering the screen
A2	User Popup	3	As a/an Admin, I want a popup of the user that I search, so that I can quickly see the information I want without other users cluttering the screen
	Total:	73	

Table 2: Functional requirements as User Stories.

3.3 AI generated stories

Given the project description and current user stories, Claude gave around 20 more stories, separated by category:

Authentication Access

- As a User, I want to reset my password, so that I can regain access to my account if I forget it.
- As an Admin, I want to assign roles to users, so that each user only has access to features relevant to their role.
- As an Admin, I want to view a list of all users and their roles, so that I can manage the team effectively.

Purchase Orders & Materials

- As a Project Manager, I want to edit or remove individual materials from a PO after it has been uploaded, so that I can correct OCR errors or changes in the order.
- As a Project Manager, I want to be notified when a delivery is flagged as incomplete or inaccurate, so that I can follow up with the supplier.
- As a Project Manager, I want to view the history of all POs associated with a project, so that I have a complete audit trail.
- As a Logistics user, I want to flag specific missing or incorrect materials on a delivery, so that discrepancies are clearly documented and actionable.
- As a Logistics user, I want to partially receive a shipment, so that I can record materials that arrived while others are still pending.

Notifications

- As a Project Manager, I want to receive a notification when a packing slip has been reviewed and compared, so that I am aware of delivery outcomes in real time.
- As a user, I want to configure which notifications I receive, so that I am not overwhelmed by irrelevant alerts.

Projects

- As a Project Manager, I want to mark a project as complete or inactive, so that it is separated from active projects without being deleted.

- As a User, I want to search or filter projects, so that I can quickly find the one I need.

Tools

- As a Foreman, I want to mark a tool as lost, damaged, or retired, so that its status is accurately reflected for all users.
- As a Foreman, I want to view the full assignment history of a tool, so that I can track its usage over time.
- As a User, I want to search for a specific tool by name or ID, so that I can quickly find its current location and status.

Serial Capture / QC

- As a Quality Control user, I want to view all serial numbers captured for a project, so that I have a complete equipment record.
- As a Quality Control user, I want to edit or correct a captured serial number, so that data entry errors can be fixed.

General / Mobile

- As a User, I want the app to work offline or in low-connectivity environments, so that I can still log deliveries on remote job sites.
- As a User, I want the interface to be optimized for mobile, so that I can easily use it on-site without a computer.

Claude also gave feedback on what was missing in our stories: "The biggest gaps I noticed were around error correction (fixing bad OCR results), partial deliveries, discrepancy flagging with clear ownership, project lifecycle management, and tool status tracking beyond just assignment. Those are likely to come up early in development so worth prioritizing."

There are clearly some mistakes. For example, the two stories under General/Mobile are better classified as NFR's. Furthermore, the assigning user roles story is already built into our A1 story. For the most part, though, these stories are good additions. If I had to pick a few to add to the project, I would add setting projects as complete/inactive and letting users reset their passwords.

4 System Design

4.1 Architecture

Our team has chosen to utilize the MVC structured architecture to allow for obvious focus point on our development. Using MVC we can split up our work into 3 main categories: Model, View, and Controller. Using this style of structure will promote scalability, mendability, and maintenance as we move further down the road. It will allow for developers to handle different parts of the project, avoiding the conflict of waiting on another group member to finish before starting your work. When it comes to tools we will use for each part of MVC that will be explained in our Technology section. The module will be the main branch that handles our data that will contain important information for the game consistently and securely. The controller is 2 the link between the model and the view. It will take in requests data from the user and retrieve data from the model to display on the view. The view is the display of the structure and it allows for the user to see the information our software is displaying. Many benefits will be created from using this structure that will allow for efficiency, debugging, organization, modular development, team collaboration, and maintainable software.

4.2 Diagrams

See figure 1 for the current class diagram. I have a few notes for the current diagram:

- Every job has one project manager and multiple logistics/foremen associated with it. I'm not sure if there's a way to differentiate them, since they are both 'Users,' so I wrote the appropriate role next to each arrow.
- The arrow pointing from Job (should be 'Project') to Material references the Material List from the purchase order. The arrow pointing from Shipment Slip to Material refers to the list of actually delivered materials.

4.3 Technology

For our project we have planned out some crucial technology that will allow us to produce a product to out client exactly as he wants. We plan to use java script for our main programming language, NodeJS for our server side run time environment, Express for the web application framework for building APIs, HTML/React/CSS for our front end framework to create a present and interactive view of the users of this app. We will focus on using react for our mobile app and HTML + CSS for our web app. For our testing we are going to use Jest to make proper tests to make sure our software work effectively. Our client asked for OCR to be used in our project to convert photos with text to database ready text. We plan to implement Tesseract.js or AWS Textract for this function of our web app. For our database we will user MongoDB to

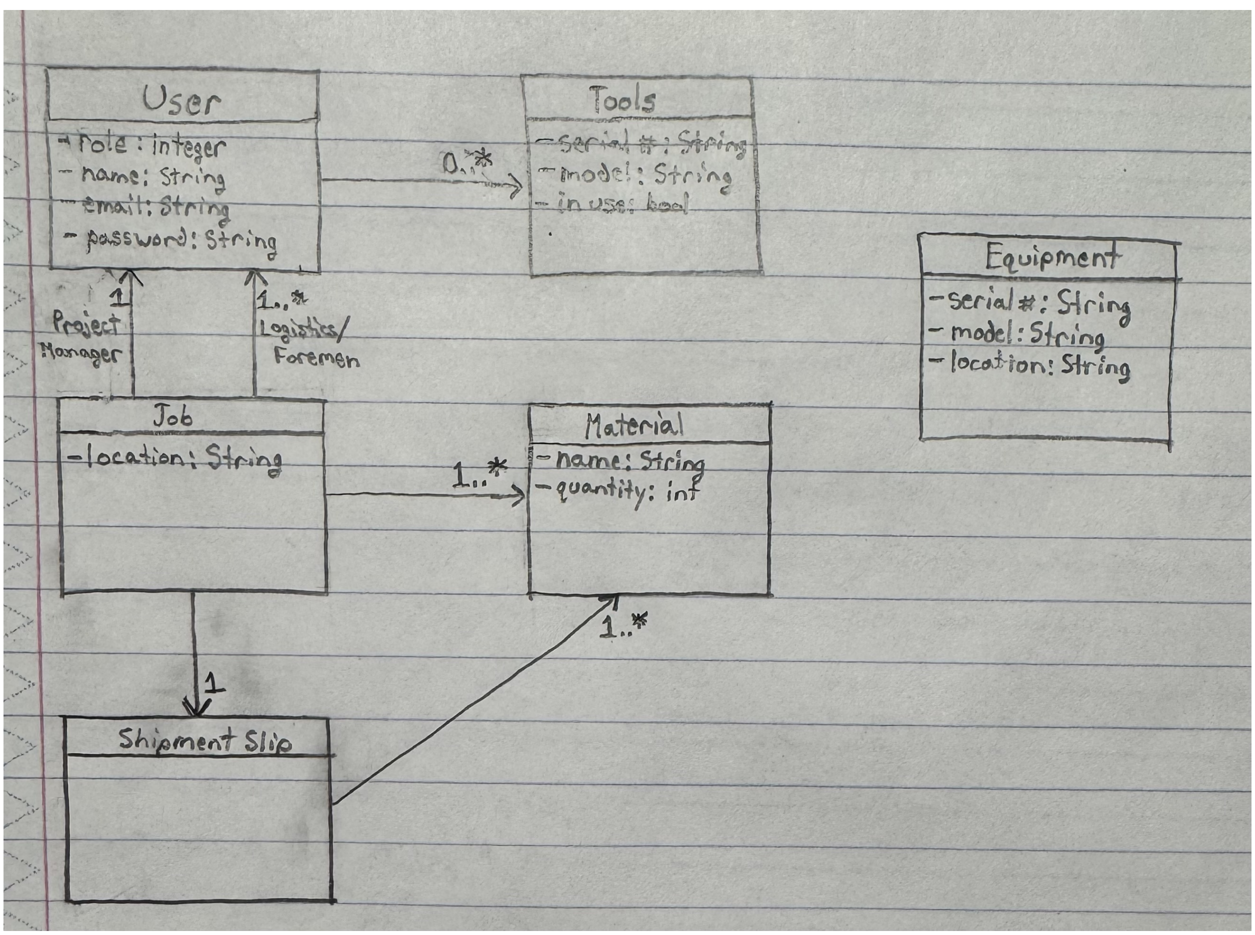


Figure 1: Current Class Diagram

store all of our information in order to produce accurate real time information to our users. Also bcrypt for hashing our passwords for better security.

4.4 Coding Standards

NoSQL data structure

Users ((_id: ObjectId, id_user: 1, username: 'sample123', password: ('123'), #hashed name: 'sample user', id_tools: (()), role: 'ProjectManager') ...) Projects ((_id: ObjectId, id_manager: 1, id_workers: (()), id_materials: (()), location: 'sample'))

Tables should have a capital first letter but following letters are lowercase

Backend

/models /controllers /routes /middleware /services /tests app.js

Consistent naming conventions

Modular architecture

Minimum 50% test coverage before a commit

Make sure both partners can agree with new code before commit to git

Secure authentication practices like hashing password and page restrictions

4.5 Data

- Material - name, quantity
- User - type (Admin, Project Manager, Logistics, Foreman), name, username/email, password, tools being used
- Tools - serial number, model, in use (bool)
- Project - location, PM, L/F on site, material list
- Equipment - serial number, model, ..., location

The '...' under the equipment data refers to attributes that I am not sure of yet. It could include different voltage levels, quality, or other things. Dealing with the equipment is still a stretch goal for now, so this is something we will worry about later.

4.6 UI Mocks

The color scheme will probably match the company colors, blue and red, though the client said that color was not a big deal.

5 Iterations

5.1 Iteration Planning

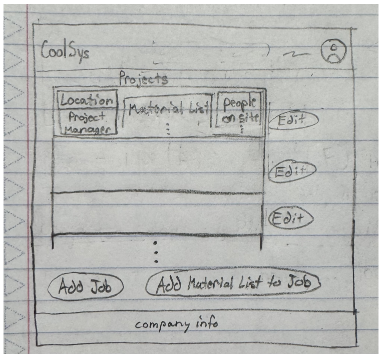
Table 3 shows the iteration planning.

[Note that the original planning was limited because we did not have all of the details for the project at the time. The specific iterations will have more stories done.]

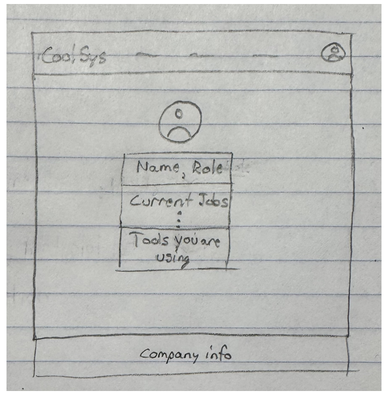
5.2 Iteration/Sprint 1

5.2.1 Planning

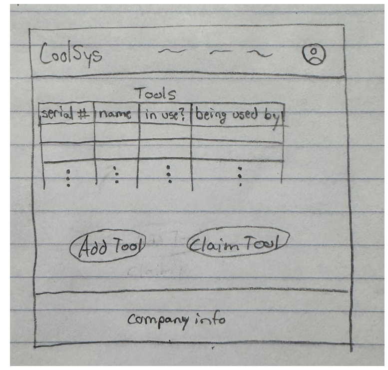
- Oscar:
  1. A1 User Accounts (8 Points)
- Dylan:



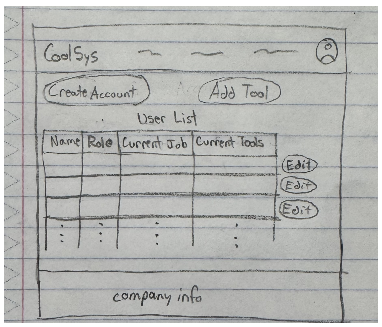
(a) Jobs Page for Project Managers



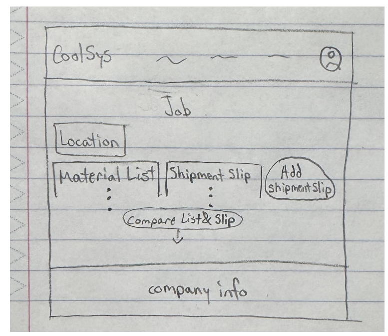
(b) User Profile Page



(c) Tools Page for Foremen



(d) Admin Page



(e) Job Page for Logistics/Foremen

Figure 2: UI Mocks

It.	Dates	Stories	Points Planned	Points Done
1	01/29 - 02/10	A1 User Accounts, P1 Material Lists, D1 Projects	16	16
2	02/01 - 03/01	D2 Organize Materials, D3 Assign Users, T1 Tool Management	13	??
3	03/01 - 04/01	D6 Assign/Unassign Tools, S1 Record Shipment, S2 Shipment Accuracy	10	??
4	04/01 - 05/01	D4 Material List Status, P2 PO Photo, S3 Shipment Photo	10+	??
5	05/01 - 06/01	D5 Shipment Notification, Q1 Project Details on Forms, Q2 Equipment Serial Capture	09	??
		Total:	58+	65

Table 3: Iteration Planning for Incremental Deliveries

1. D1 Projects P1 Material list (8 points)

- Total Points Planned: 8+

For the first iteration, we focused on setting up the essential DAO's and their controllers, those being users and projects. Also, setting up basic UI should try to get done to keep frontend and backend on the same track.

```
File                             | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
---------------------------------|---------|----------|---------|---------|-------------------
All files                        |   95.12 |       75 |   96.29 |   95.08 |
 CS-Senior-Capstone              |    90.9 |       50 |      75 |      90 |
  dbconnection.js                |    90.9 |       50 |      75 |      90 | 9
 CS-Senior-Capstone/controller   |   92.06 |      100 |     100 |   92.06 |
  ProjectController.js           |     100 |      100 |     100 |     100 |
  UserController.js              |   83.87 |      100 |     100 |   83.87 | 29-31,41,53
 CS-Senior-Capstone/model        |     100 |      100 |     100 |     100 |
  ProjectDao.js                  |     100 |      100 |     100 |     100 |
  UserDao.js                     |     100 |      100 |     100 |     100 |
---------------------------------|---------|----------|---------|---------|-------------------
Test Suites: 6 failed, 4 passed, 10 total
Tests:       27 passed, 27 total
Snapshots:   0 total
Time:        4.201 s
```

Figure 3: Iteration 1 test coverage report

5.2.2 Work Done

- Oscar:
    1. A1 User Accounts (8 Points)
- Dylan:
    1. D1 Projects (8 points)
- Total Points Done: 8+

Oscar: I completed model and controller for users, which includes full CRUD functionality. Basic UI was also set up for the users, including a form to register and a list of users. While the UI is simple, there are still improvements that can be made on the style.

Dylan: I had planned to most of projects crud and and a portion of material list but decided to just complete the full crud for Projects. I have created two UIs which was one to handle create view and delete but also added a button to make updates to a desired project. I still see room for alot of improvement in how users are added to projects.

5.2.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

I think our coverage is done very well because it reaches close to perfect coverage in all fields. I think there is room for very little improvements just because we are almost near perfection. User controller test is missing some lines just because they arent very important for testing but in time could easily be covered in the next iteration. This is the first iteration of testing and cant be compared to anything in previous iterations.

5.2.4 Retroespective & Reflection

Oscar: The only thing that I wasn't able to do this iteration was the testing for the functions in the script on the users admin page. I will have to make that up eventually. However, both the user DAO and controller files are above 80% coverage, so I am still confident in the user functionality on the website. I also did some work on working with Bootstrap. While I used it in my software engineering project, I mostly copied from my teammates and did not really learn it. Learning Bootstrap will be something that I build on throughtout this project.

Dylan: When it comes to how well the usability of the projects I think it could a little work on selecting users. I ran out of time and just created a simple method of just adding the user ID manually instead of having a drop down. I belive once the drop down is added I will not have to add much work the the project crud. I think I could have spent more time researching how we could possibly use a camera to input information from documents because I think this will be a very strenuous problem later on.

5.3 Iteration/Sprint 2

5.3.1 Planning

- D2 Organize Materials (3 points)
- P1 Material Lists (3 Points) may be 5
- D3 Assign Users (2 points)
- T1 Tool Management (8 points)
- M1 Persistent Login/Logout (2 points)
- Total points: 18

Now that users and projects are set up, the next step is to link them by assigning users to projects. Other things that we are starting are setting up tools and adding login/logout (which also means hashing passwords and setting access permissions by role). While the total points at the time this is written (15) is low, M1 might end up being worth more points, and D2 might be combined with another story. Since we have created a full crud for materials[8] we need to also create the full crud for shipment as well. Once shipment is complete we will begin to implement the comparing feature that will compare a material list to a shipment then update its status.

5.3.2 Work Done

- Oscar:
    1. T1 Tool Management (8 points)
    2. M1 Persistent Login/Logout (2 points)
    3. part of (D6) Assign/Unassign Tools (1 point)
- Dylan:
    1. D2 Organize materials (3 points)
    2. P1 Material Lists (3 or 5 points)
- Total Points Done: 18+

Dylan: For this sprint it was important that we created the create of crud for materials to make sure that projects could have a material list in order to make sure project managers can create and also foreman can view. During implementation i realized that installing the full crud would be a good idea. I needed to change a user story from organize the materials to just a full crud version. After creating the create function I went ahead an implemented all the other features including update and delete, completing the CRUD. **Oscar:** I was mostly successful in this sprint. Login/Logout is totally done, which includes sessions and encrypted passwords. The Tool Management is very close to being done. Finally, I have added functionality to assign users to tools, though it might make more sense to have it the other way around. I also made small improvements and additions to the User stuff.

In general, functionality for users and tools is up on the website. The only issue is an unknown error when trying to update anything. Even though an error shows up, the update still goes through. This is something that I will keep working on, but I am not too worried about right now because there is still more functionality to add.

```
-------------------------------|---------|----------|---------|---------|---------------------------
File                           | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------------|---------|----------|---------|---------|---------------------------
All files                      |   92.45 |    72.72 |   98.11 |    92.3 |
 CS-Senior-Capstone            |    90.9 |       50 |      75 |      90 |
  dbconnection.js              |    90.9 |       50 |      75 |      90 | 9
 CS-Senior-Capstone/controller |   87.74 |       75 |     100 |   87.41 |
  MaterialController.js        |   73.52 |       50 |     100 |   73.52 | 26-27,36,42-43,57-58,71-72
  ProjectController.js         |     100 |      100 |     100 |     100 |
  ToolController.js            |   85.71 |       50 |     100 |   83.87 | 21-23,33,51
  UserController.js            |   90.74 |      100 |     100 |   90.74 | 35-37,47,59
 CS-Senior-Capstone/model      |     100 |      100 |     100 |     100 |
  MaterialDao.js               |     100 |      100 |     100 |     100 |
  ProjectDao.js                |     100 |      100 |     100 |     100 |
  ToolDao.js                   |     100 |      100 |     100 |     100 |
  UserDao.js                   |     100 |      100 |     100 |     100 |
 CS-Senior-Capstone/util       |     100 |      100 |     100 |     100 |
  Hashing.js                   |     100 |      100 |     100 |     100 |
-------------------------------|---------|----------|---------|---------|---------------------------
Test Suites: 4 failed, 9 passed, 13 total
Tests:       54 passed, 54 total
Snapshots:   0 total
Time:        7.969 s, estimated 9 s
Ran all test suites.
```

Figure 4: Iteration 2 Coverage

5.3.3 Testing Coverage

The testing for our project we would say for our project is up to standard. Most of our functions achieve good branch and coverage function but could use a little work to get perfection. In order to improve our coverage, we should make sure that error handling is tested because that is where we lack certain function coverage. Also, for the Material controller, we noticed problems with conflicts with other testing since it contains User and Projects as attributes. So finding a fix to that should also ensure better coverage. Most of the testing has stayed the same and used similar patterns to previous testing, but adapted to handle new functions.

5.3.4 Retroespective & Reflection

Dylan: Some of the challanges I faced was dealing with matrial list's inclusion of users and projects. It was a bit of a learning curve dealing with interactions between classes but once I got it down it all worked out well. Another problem was testing because we mentioned previously, testing the material controller caused problems that I think were due to the tests being run all at once. I removed certain testing to fix it, but I believe a test runInBand might work. This process of iteration two went pretty well since it was very similar to the first but once I begin to integrate shipments and comparing for validation, I expect to run into some problems. I would say a valuable thing I learned this iteration is how to make multiple classes all work with each other. It was something that was a bit out of my mind until now, but I found that using an observer-type method allowed for a change in one place to affect all the others.

Oscar: I didn't have trouble with adding any of the functionality this iteration, but I am not set on the visuals yet. There is definitely more I can do to make it look better. However, the combo of not being a very artistic person and the need to prioritize makes it tough to try. I hope that I can add to the design little by little as the project goes on, but that will only happen if I put some effort into it. Also, so far I have been working on stuff that I am comfortable with, but soon we will start getting into unfamiliar territory, especially with the OCR, so there will be more trial and error with the process down the road. Finally, I met with the client last week and asked for the PO and shipment slips we will need to base our materials (and OCR) off of. It hasn't been necessary yet, so it is not a big deal, but it will be needed soon.

5.4 Iteration/Sprint 3

5.4.1 Planning

- (the rest of) D6 Assign/Unassign Tools (2 points left)
- S1 Record Shipment (2 points)
- S2 Shipment Accuracy (5 points)
- D8 User Dashboard (5 points)
- Total points: 14

5.4.2 Work Done

- Oscar:
  1. (the rest of) D6 Assign/Unassign Tools (2 points)
  2. D8 User Dashboard (3 points, previously 5)
  3. (Spike) UI Change (2 points)
- Dylan:
  1. S1 Record Shipment (8 points(Full Crud), previously 2(Just create function)
  2. S2 Shipment Accuarcy (2 points partial of the full 5 points)
- Total Points Done: 17

5.4.3 Testing Coverage

Oscar: I was able to finish the tests for error handling, so all of the code that I have written, aside from the js files accompanying the html pages, is at minimum 97% coverage.

5.4.4 Retroespective & Reflection

Oscar: I did not get as much done as I wanted to this sprint. I was at a tournament in South Carolina for most of spring break, so I was unable to do any work for about a week. However, there was time before my trip that I should have done more work. I finished the assigning/unassigning tools story, though more might get added to it depending on if we decide to change the data display on the website. I finished the User Dashboard, but it only ended up being worth 3 points right now. If anything else is added to it, extra points might be added as an additional story. I also put in a lot of time (hopefully a spike's worth) into fixing the design of the website, and I hope that most of the changes will be permanent. I also added error handling tests to the DAO's and controllers that I wrote, which is something that I had to catch up on. Overall, I thought I did okay this sprint given the time constraint, but I will need to pick up the pace for the remaining sprints.

Dylan: This sprint was definitly the hardest sprint out of all of them because I found that I was tiring myself out on the front-end aspect of Shipments. After receiving feedback on our front end I went to find out how to make better ways of selecting instead of a dropdown. I tried multiple methods but finally found one that works best. Involves populating a drop down based off what the user types in a text box. After this I ran into a issue that involes our planning structure, which was how we store our materials. Once I finished shipment and tried to incorporate material checks, I realized that a Project should have Orders and the Orders should have Material lists. This way when creating a shipment you can just select a Project then that Project populates a Orders and then you can enter the received materials for proper Shipment accuracy checking. I ran into a lot of problems that stopped a lot of progress, so i decided to just focus on completing the full crud of Shipment instead of doing the shipment accuracy. I was still able to a bit of shipment accuracy adding to about 2 points worth.

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	94.62	78.94	97.05	94.75	
CS-Senior-Capstone	90.9	50	75	90	
dbconnection.js	90.9	50	75	90	9
CS-Senior-Capstone/controller	92.68	80.55	100	92.92	
MaterialController.js	80.64	66.66	100	80.64	10-16,29-30,38-39
ProjectController.js	90.32	100	100	90.32	9-16
ShipmentController.js	88.63	80	100	88.09	9-14,27-32
ToolController.js	97.61	70	100	100	40-41,43
UserController.js	100	100	100	100	
CS-Senior-Capstone/model	98.21	100	97.05	98.16	
MaterialDao.js	100	100	100	100	
ProjectDao.js	94.11	100	85.71	92.85	37
ShipmentDao.js	95.23	100	100	95.23	21
ToolDao.js	100	100	100	100	
UserDao.js	100	100	100	100	
CS-Senior-Capstone/util	100	100	100	100	
Hashing.js	100	100	100	100	

```
Test Suites: 8 failed, 5 passed, 13 total
Tests:       14 failed, 71 passed, 85 total
Snapshots:   0 total
Time:        3.156 s
Ran all test suites.
```

Figure 5: Iteration 3 Coverage

5.5 Iteration/Sprint 4

5.5.1 Planning

- P3 Project Popup (3 points)
- T2 Tool Popup (3 Points)
- A2 User Popup (3 points)
- S1 Record Shipment (2 points) [adjustments]
- S2 Shipment Accuracy (3 points)
- D2 Organize Materials (3 points)
- P1 Material Lists (5 points)

5.5.2 Work Done

- Oscar:
  1. P3 Project Popup (3 points)
  2. T2 Tool Popup (3 points)
  3. A2 User Popup (3 points)
  4. Fixed error on updating
- Dylan:
  1. S1 Record Shipment (2 points) [adjustments]
  2. S2 Shipment Accuracy (3 points)
  3. D2 Organize Materials (3 points)

PASS util/Hashing.test.js

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	99.1	79.41	97.1	99.37	
CS-Senior-Capstone	90.9	50	75	90	
dbconnection.js	90.9	50	75	90	9
CS-Senior-Capstone/controller	99.53	81.25	100	100	
ProjectController.js	100	100	100	100	
PurchaseOrderController.js	100	100	100	100	
ShipmentController.js	100	75	100	100	8-12
ToolController.js	98.03	70	100	100	40-41,43
UserController.js	100	100	100	100	
CS-Senior-Capstone/model	99.02	100	97.05	98.96	
ProjectDao.js	94.11	100	85.71	92.85	33
PurchaseOrderDao.js	100	100	100	100	
ShipmentDao.js	100	100	100	100	
ToolDao.js	100	100	100	100	
UserDao.js	100	100	100	100	
CS-Senior-Capstone/util	100	100	100	100	
Hashing.js	100	100	100	100	

Test Suites: 11 passed, 11 total
Tests: 95 passed, 95 total

Figure 6: Iteration 4 Coverage

4. P1 Material Lists (5 points)

- Total Points Done: 22

5.5.3 Testing Coverage

Testing for this iteration went very well, and we were able to make some improvements that made our coverage a lot better. First, we had some failed tests, but after using –runInBand it fixed all our issues, and all tests passed. Tesing for Matrials was removed since it was not needed, and testing for Purchase Order was added. We still have a little imperfections but overall the testing came out very well and we are proud of our current standings.

5.5.4 Retroespective & Reflection

Oscar: This sprint did not go as originally planned for me. Our original idea for displaying all of the data was in tables. However, we underestimated the amount of stuff we would need to display. There are going to be hundreds of projects and thousands of tools active at the same time, which would not work in a table format. Because of this, we had to come up with a new idea. Dylan added a search feature that searches by an attribute (email for users, serial number for tools, etc), and I added a popup with all of the data's information. Even though this was a new user story, there was no new functionality added. It also took me a little over a day to solve the bug that popped up when updating stuff. Going into the final iteration, most of the functionality that we wanted to add has been done. However, most of it is in the context of admin control. We will need to add less exhaustive features for the other roles.

Dylan: This sprint was pretty much just making adjustments after the client revision. I had originally meant to just make all the UI of my cruds better fit for mass amounts of data and also look better. Although after being told the material list was not exactly what he wanted, I had to completely delete Materials Crud and restart with a Purchase Order Crud. This took alot of work because it was making a full new CRUD but also changing every other class that it affected. After I created the implementation of Purchase Order,s I was finally able to do the Shipment accuracy, so when logging a shipment, it would check for accuracy with the corresponding Purchase Order. This was a very frustrating Iteration just because I made a mistake and had to delete a lot of code from iteration 2.

5.6 Iteration/Sprint 5

5.6.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project. Be concrete about your progress, you know how many story points your software is, how many points you completed (this shows your progress). You also how many points your team delivers at each iteration, therefore you can estimate how many more iterations it would take to finish the leftover points (show the math).]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently? How can you do better work in future projects? You may write this as a team or per person (or both — if all your iterations were team reflections, then it would be better to write individual reflections here)]

Appendix

[Appendix section if needed]